

yet versatile, text editor for Debian Linux. Close the current window and open a terminal to write the following commands:

```
pi@raspberrypi:/$ sudo nano /etc/webcam.conf
[ftp]
host = localhost
dir = /var/www
file = webcam.jpg
tmp = imageup.jpg
local = 1
[grub]
device = /dev/video0
width = 176
height = 144
delay = 5
quality = 75
trigger = 180
```

Save the file with `ctrl+o` command. Exit Nano using `ctrl+x` command. Close it and go back to the terminal window.

Note that, the webcam has a versatile software, with an ability to continuously ftp the captured image to a remote server, where the main webpage may be the host. But, in this case, we have made `local = 1`, which means that we will host it on Raspberry Pi's Apache server page (`/var/www` directory) itself.

The `delay = 0` means the cam will take pictures non-stop. Since this will put a lot of load on the tiny Raspberry Pi board, a delay of five (means, taking one snap every five seconds) will be more justified for Raspberry Pi.

Checking the working of Apache server. We will now check whether Apache web server is working or not. In the Raspberry Pi computer, open a Midori browser page and enter the address as:

```
http://localhost
```

Apache will open a page, indicating the server is up-and-running (even though there is no content page available in the server). So far, so good. Now, we have to see whether the same page is available over wireless or not.

Before that, we have to know the IP address of our Raspberry Pi board

(mine is a DHCP by my wireless router).

Issue the following command at the terminal window:

```
pi@raspberrypi:/# ifconfig
```

This will give details of the IP addresses of the wired as well as the wireless network of Raspberry Pi board. Look out for WLAN configuration—WLAN IP address of my Raspberry Pi board as assigned by my router is 192.168.1.2. So, I enter the following address in the address bar of the browser on my laptop:

```
http://192.168.1.2/
```

The same page opens up in my laptop. This confirms that Apache is up-and-running on Raspberry Pi board.

Automatic capturing of images.

Open the terminal window and enter the following command:

```
pi@raspberrypi:/# sudo webcam /etc/webcam.conf &
```

The last `&` command means the process will execute in the background. Now, go back to a remote laptop and enter the following address in the browser address bar (To stop the webcam, use `ctrl+c` command or find out process ID number and then kill it by `$sudo kill-9xxxx`, where `xxxx` is the process number.):

```
http://192.168.1.2/webcam.jpg
```

A still-image facing the camera will now appear on the top-left corner of your laptop. This confirms that the webcam is working. You may try to open the same page in Midori browser on Raspberry Pi, but it may take several minutes to open, as most of its resources are busy handling webcam images. Do not worry; the page will open on the remote laptop quite comfortably.

Change the direction of the camera and refresh the browser on your laptop, and the image will change. Unless the browser window is refreshed, the picture will not change.

Now, we will create an automated page-refresh in Raspberry Pi itself.

Automated page-refresh from Raspberry Pi. Let us now use Nano to create a simple html page in

`/var/www` directory of Raspberry Pi. Note that, `/var/www` is an Apache document root directory. Whatever we write here will be available, to view, from any PC connected to the network.

```
pi@raspberrypi:/# sudo nano /var/www/webcam.html
<html>
<head>
<META HTTP-EQUIV="pragma" CONTENT="no-cache">
<META HTTP-EQUIV="refresh" CONTENT="0">
</head>
<body>
<center>
<h2> My Raspi Webcam </h2>

</center>
</body>
</html>
```

Save using `ctrl+o` and exit using `ctrl+x` keys on the keyboard. Now, go to the remote laptop and open the page `http://192.168.1.2/webcam.html`

The image will be auto-refreshing every second now. But, this may be a little irritating to the eyes, so we may need to enter a delay factor here. In the webcam, `html CONTENT="0"` can be made to `CONTENT="5"` for a more soothing refresh once in every five seconds.

You can strip Raspberry Pi board from the mouse, keyboard and monitor, and the webcam will still work non-stop—such is the power of Raspberry Pi and Linux.

Complete automation. Now, how about an automated script? So that, every time you boot or even before login, Raspberry Pi will fire-up the webcam command.

You can get rid of the monitor, keyboard and mouse from Raspberry Pi board. For that, we will put the command in `/etc/rc.local` file, and every time, on boot up, the webcam will start. Open a shell script file `webcam.sh` in `/home/pi` and enter the following command:

```
pi@raspberrypi:/# sudo nano /home/pi/webcam.sh
```